D Y PATIL COLLEGE of ENGINEERING & TECHNOLOGY (AN AUTONOMOUS INSTITUTE) KASABA BAWADA, KOLHAPUR

VisionShield – AI-Based Foundry Monitoring System

Real-time sensor monitoring · AI PPE safety detection · SOP process verification

Group G49 Snehal S. Wagavekar · Sanika V. Gurav · Shweta R. Vhannure · Sufiya B. Bargir · Samiksha V. Gaikwad

Guide: **Mr. Niraj A. Haval** | B.Tech CSE (Final Year) | 2026–27

1 Abstract

VisionShield monitors foundry operations with computer vision and sensor data. A YOLOv8 model detects workers and protective equipment, three trained process models verify that the casting steps follow the SOP – including the mandatory magnesium treatment – and the system raises alerts when temperature or weight leaves the safe range. Everything is available on a role-based web dashboard backed by MongoDB.

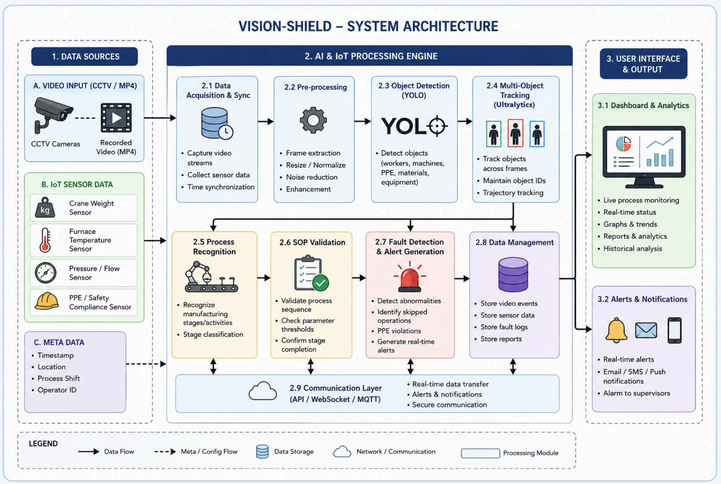
2 Problem Statement

- Manual monitoring is slow and prone to human error
- Skipped or wrong process steps reduce product quality
- Safety violations (missing PPE) can go unnoticed
- No real-time alerts or traceability for root-cause analysis

3 Objectives

- Monitor the foundry process automatically
- Detect PPE compliance and process abnormalities in real time
- Verify the SOP step sequence
- Generate real-time alerts, reports and analytics
- Improve product quality and worker safety

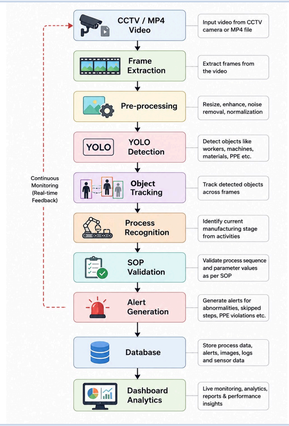
4 System Architecture



VisionShield system architecture

5 Five Integrated Modules

- **Module 1 : Video Acquisition & Data Collection**
- **Module 2 : AI-Based Object Detection & Tracking**
- **Module 3 : Manufacturing Process Recognition & SOP Validation**
- **Module 4 : Fault Detection & Alert Generation**
- **Module 5 : Database Management & Analytics Dashboard**



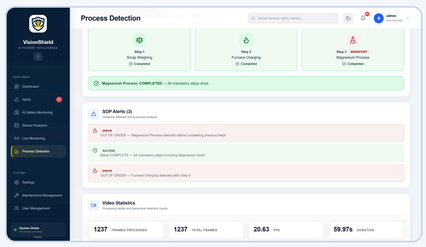
Processing workflow

How it works

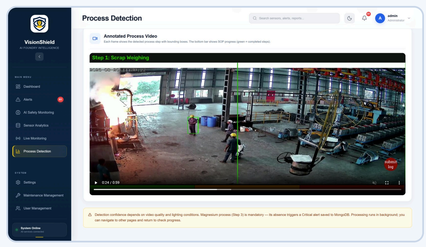
- Video and sensor data are collected
- YOLOv8 detects people and PPE; tracking keeps object IDs
- Three models recognise process steps: Scrap Weighing, Furnace Charging, Magnesium
- SOP order and mandatory steps are validated
- Faults raise alerts, stored in MongoDB and sent by e-mail

6 Results

1,237	3	10	24/32
frames analysed in the test process video	SOP models run together	PPE detections on the test image	test cases passed, 0 failed

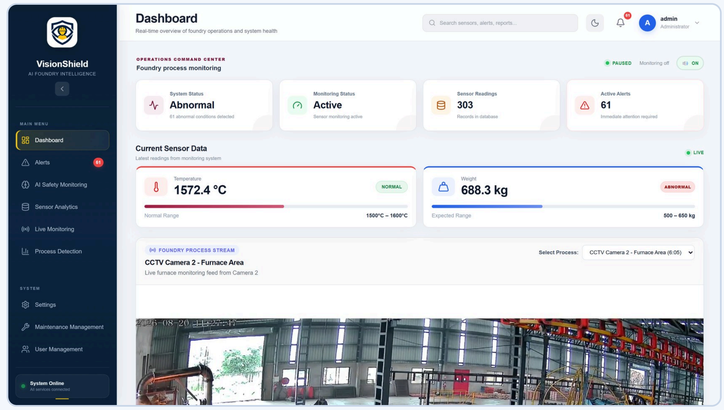


SOP sequence verified from video

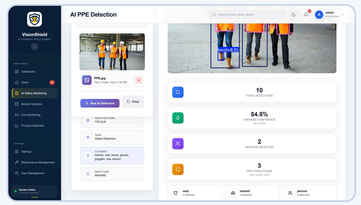


Annotated output video

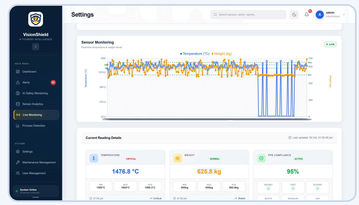
7 Working System



Dashboard – status, sensor readings and alerts



AI PPE detection



Live monitoring charts

8 Technology

React + Vite · FastAPI · MongoDB Atlas · YOLOv8 (Ultralytics) · OpenCV · PyTorch · Python 3.13

- Role-based access: Administrator, Supervisor, Operator
- E-mail alerts, equipment maintenance and user management

9 Conclusion

All five modules are integrated into a working web application and were tested with real foundry footage. Future work: live RTSP cameras, ESP32 sensors, model accuracy evaluation (mAP) and larger real-factory datasets.

10 Source Code & Demo

Scan for the GitHub repository

github.com/bargirsufiya/VisionShield